

# **JNTU Kakinada**

## **REGULAR EXAMINATION ? 2024**

Network Analysis ? Regular Examination, 2024

Subject Code: EC304 Semester: 3 Regulation: R23

Time: 3 Hours Max. Marks: 70

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Note: This is a demo question paper generated for Backlog Buddy.

Answer all questions. Assume suitable data wherever necessary.  
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### **SECTION A (Unit 1: Mesh and Nodal Analysis)**

1. Define Mesh and Nodal Analysis. Explain its significance in Network Analysis with a suitable example. [12 Marks]
2. Discuss the engineering applications of Mesh and Nodal Analysis. [12 Marks]
3. Describe the working principle of Mesh and Nodal Analysis with necessary equations. [12 Marks]

### **SECTION B (Unit 2: Network Theorems)**

4. Write short notes on Network Theorems. [12 Marks]
5. List the advantages and limitations of Network Theorems. Explain any two in detail. [12 Marks]
6. How is Network Theorems implemented? Give an algorithm / procedure and explain. [12 Marks]

### **SECTION C (Unit 3: Two-Port Networks)**

7. Compare and contrast the key concepts involved in Two-Port Networks. [12 Marks]
8. Derive the expression related to Two-Port Networks and explain each term. [8 Marks]
9. How is Two-Port Networks implemented? Give an algorithm / procedure and explain. [12 Marks]

### **SECTION D (Unit 4: Resonance)**

10. Solve a problem related to Resonance. Show all the steps clearly. [8 Marks]
11. Describe the working principle of Resonance with necessary equations. [12 Marks]
12. How is Resonance implemented? Give an algorithm / procedure and explain. [12 Marks]

### **SECTION E (Unit 5: Transient Analysis)**

13. Discuss the engineering applications of Transient Analysis. [12 Marks]
14. Describe the working principle of Transient Analysis with necessary equations. [12 Marks]
15. Solve a problem related to Transient Analysis. Show all the steps clearly. [8 Marks]

**\*\*\* End of Question Paper \*\*\***